complemented by a new generation of enormous terrestrial telescopes, flooded cosmologists with data. Around the same time, the advent of high-speed computing made it possible not only to process this data but to simulate the entire universe, testing out different theories about its composition and behavior. Cosmology rapidly went from guessing some of the most fundamental properties of the universe to nailing them down with shocking precision. In 1996, estimates for the age of the universe ranged from 10 to 20 billion years, much as they had for the three decades since Penzias and Wilson had discovered the CMB. By 2006, the age of the universe had been pinned down to 13.8 billion years, give or take a percent.

With that precision came new pictures of the universe. The Wilkinson Microwave Anisotropy Probe (WMAP), a space telescope launched in 2000, created a detailed map of the tiny differences in intensity in the CMB, which is uniform to about one part in 100,000. That map lent support to a theory of the extremely early universe at the time of the Big Bang, known as "inflation." Inflation, first proposed by the physicist Alan Guth in 1981 and refined by Andreas Albrecht and Andrei Linde shortly thereafter, says that the very early universe expanded extraordinarily quickly for a minuscule fraction of a second—increasing in size by a factor of about 100 trillion trillion in about a billionth of a trillionth of a trillionth of a second—then resumed expanding much more slowly. This rapid expansion was driven by hypothetical "inflatons," high-energy subatomic particles, which decayed into normal matter at the end of inflation. Crucially, the theory dictated that tiny quantum fluctuations in the density of inflatons were blown up by the inflation process, and then led to the tiny fluctuations in the density of normal matter in the small, hot universe immediately after inflation. Those fluctuations led, in turn, to the fluctuations in the CMB—and ultimately seeded the formation of all structure in the universe today, including our own galaxy and Earth itself. In short, inflation suggested that we are all the products of quantum fluctuations that occurred in the very early universe—and WMAP's data suggested inflation was right. "WMAP's data supports the notion that galaxies are nothing but quantum mechanics writ large across the sky," Brian Greene said in 2006. "This is one of the marvels of the modern scientific age."

But the Copenhagen interpretation couldn't explain how the early universe worked—and the mathematics of quantum physics couldn't handle